

PARTICIPANT INFORMATION SHEET

Study title

Gait Recognition-based Authentication for User-Independent Physical Access Control

Brief summary

This project explores the development and feasibility of a gait recognition-based authentication system for physical access control, aiming to provide a less obtrusive and more user-independent alternative to traditional biometric and token-based systems.

Who is doing the research and why, and what is involved in the overall study?

I, Rafferty Simms, am a final-year Computer Systems (Cyber Security) Undergraduate at Nottingham Trent University, conducting my final year project on assessing how gait recognition could be implemented within a physical access control system.

Participation in this study is entirely voluntary, and you are able to stop participating at any point without giving reason. Biometric information collected during these tests will be pseudo-anonymised and stored securely following university guidelines and UK legislation for the duration of testing, before being securely disposed of when no longer needed.

What would taking part involve for the participant, and what is the time commitment?

If participating in these tests, you will be expected to 1: walk through a camera's field of view in order to allow your walking style to be captured, and 2: fill out a questionnaire detailing your experience during testing and demonstration of the system using the biometric data you provided. The expected testing time per-subject is 10 minutes, with the questionnaire taking an extra 5-10 minutes depending on the amount of feedback you wish to give.

What are the possible benefits and disadvantages / risks of taking part?

There are no direct or immediate benefits to anyone but the researcher who benefits from the results of these tests as they enable evaluation of his solution.

However, research in this area may lead to the acceptance and adoption of frictionless access control in real-world scenarios in the future – a convenience benefit for society.

A risk assessment has been conducted on the topic of this research and content of these tests, with risks to volunteers found to be negligible.

What if something goes wrong?

Although unlikely, if something does go wrong during testing or in following stages of evaluation with effect to volunteers, the supervising academic member of staff: Dr. Nemitari Ajenka (nemitari.ajienka@ntu.ac.uk) should be contacted directly to raise concerns or request additional information.

What will happen if I don't want to carry on with the study?

As previously mentioned, participation in this study is entirely voluntary, and you are able to stop participating at any point without giving reason.

If you would like to withdraw from this research at any point, having pseudo-anonymised data (if not yet deleted) or any submitted questionnaires disposed of, please email me (rafferty.simms2021@ntu.ac.uk) or Dr. Nemitari Ajenka (nemitari.ajienka@ntu.ac.uk). As results will be used to provide insights into the performance, accuracy, and acceptability of the system, complete withdrawal after data analysis has begun may not be possible.

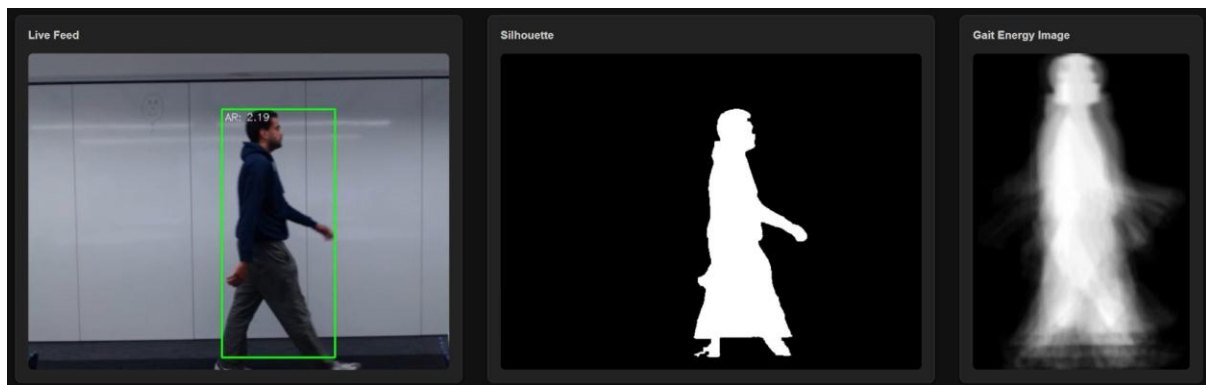
How will my information be kept confidential?

Research is a task that we perform in the public interest. Nottingham Trent University is the data controller. This means that we, as Nottingham Trent University researchers, are responsible for looking after your information and using it properly. We will use the minimum personally identifiable information possible and perform pseudo-anonymisation – samples will be labelled by unique codes. This will be stored safely in NTU data storage provision, with all biometric data destroyed following the conclusion of testing.

The age and gender of the subject will be recorded, for reference when discussing the demographic split of the dataset, as these experiments target diversity to reduce bias.

Further supporting information (full guidance covering further supporting information)

Below is an example of how your biometric gait information will be collected and represented:



Version control

Document Version 2, February 3rd, 2025

Contacts

For additional information, or to make a query please email me (rafferty.simms2021@ntu.ac.uk) or Dr. Nemitari Ajenka (nemitari.ajienka@ntu.ac.uk).